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LAB NO 05

Operating System Lab

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LAB TASK

Lab Activity 1 — Student Result & Scholarship Eligibility

Problem Statement

Solve a student result and scholarship eligibility problem by developing a shell script that takes the student's name, marks, and attendance percentage as input.

The script should:

- **Use** conditional statements to determine and display the student's grade according to the marks.
- **Show** Pass if marks are 50 or above; otherwise, show Fail.
- **Use** multiple conditions to determine scholarship eligibility.
- **Display** Eligible for Scholarship if marks are 85 or above **and** attendance is 90% or above.
- **Report** Not Eligible for Scholarship if the eligibility criteria are not satisfied.

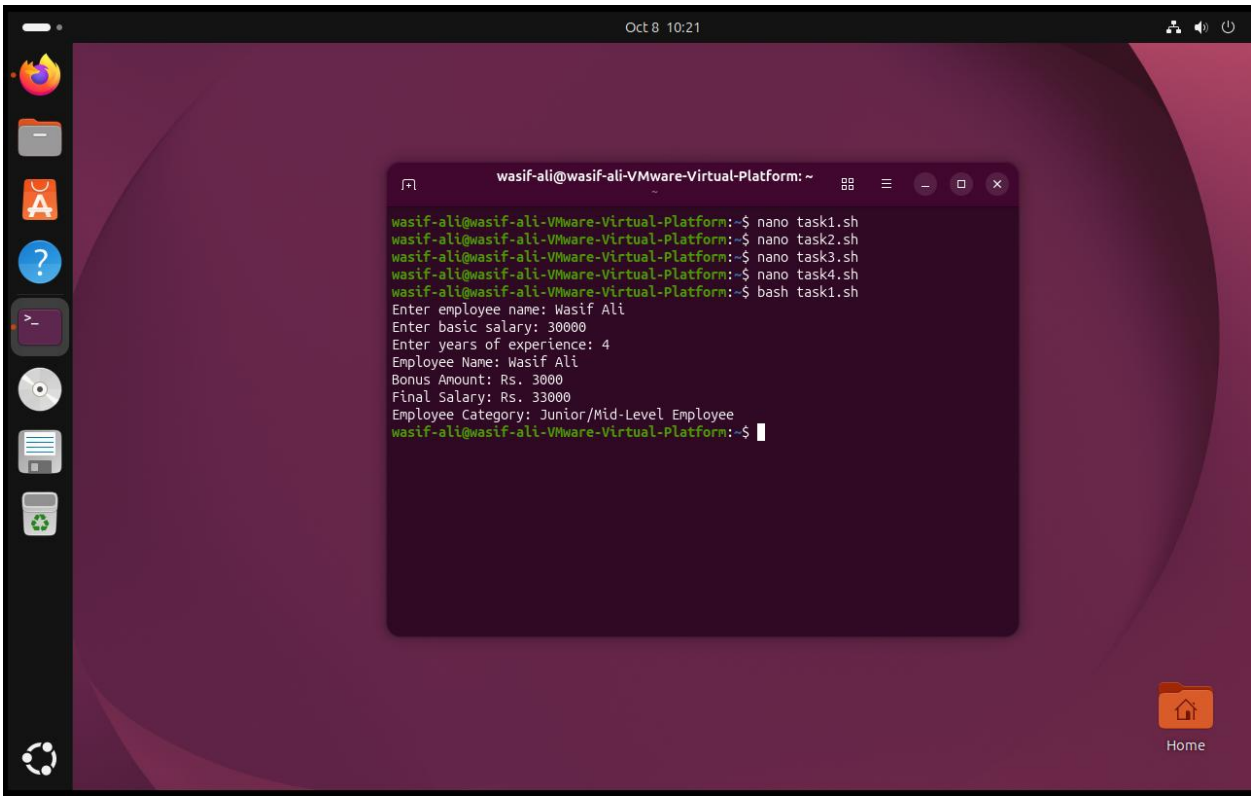
Lab Activity 1: Student Result & Scholarship Eligibility

- The script takes a student's name, marks, and attendance percentage as input.
- It determines whether the student passes (marks of 50 or above) or fails.
- It assigns a letter grade based on the entered marks.
- It evaluates scholarship eligibility, requiring both marks of 85 or above and an attendance rate of 90% or above.

Script File (Shell Script Code)**:**



Commands:



Lab Activity 2 — Employee Salary Category

Problem Statement

Prepare a shell script that takes an employee's name, basic salary, and years of experience as input and **calculates** the employee's final salary and category.

The script should:

- **Use** conditional statements to determine the bonus according to experience:
  - 5 years or more → 20% bonus
  - 3–4 years → 10% bonus
  - Less than 3 years → 5% bonus
- **Calculate** and **show** the bonus amount.
- **Calculate** and **display** the final salary after adding the bonus.
- **Relate** years of experience with the appropriate employee category.
- **Show** Senior Employee if experience is 5 years or more.
- **Report** Junior/Mid-Level Employee otherwise.
- **Demonstrate** the use of if-elif-else for multiple conditions.

Lab Activity 2: Employee Salary Category

- The script takes an employee's name, basic salary, and years of experience as input.
- It calculates a bonus based on experience tiers: 20% for 5 or more years, 10% for 3 to 4 years, and 5% for less than 3 years.
- It outputs the final salary by adding the calculated bonus to the basic salary.
- It categorizes the employee as a "Senior Employee" (5 or more years of experience) or "Junior/Mid-Level Employee" (under 5 years).

Script File (Shell Script Code)**:**

```
wasif-ali@wasif-ali-VMware-Virtual-Platform: ~ — nano task2.sh
  GNU nano 8.7.1                                  task2.sh
#!/bin/bash
read -p "Enter employee name: " name
read -p "Enter basic salary: " salary
read -p "Enter years of experience: " exp

# Calculate Bonus and Employee Category
if [ $exp -ge 5 ]
then
    bonus=$((salary * 20 / 100))
    category="Senior Employee"
elif [ $exp -ge 3 ] && [ $exp -le 4 ]
then
    bonus=$((salary * 10 / 100))
    category="Junior/Mid-Level Employee"
else
    bonus=$((salary * 5 / 100))
    category="Junior/Mid-Level Employee"
fi

Floppy Disk=$((salary + bonus))

echo "Employee Name: $name"
echo "Bonus Amount: Rs. $bonus"
echo "Final Salary: Rs. $final_salary"
echo "Employee Category: $category"
```

Commands:

```
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$ bash task2.sh
Enter employee name: Wasif
Enter basic salary: 30000
Enter years of experience: 4
Employee Name: Wasif
Bonus Amount: Rs. 3000
Final Salary: Rs. 33000
Employee Category: Junior/Mid-Level Employee
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$
```

Lab Activity 3 — ATM Transaction Validation

Problem Statement

Solve an ATM transaction validation problem by developing a shell script that takes the account balance and withdrawal amount as input.

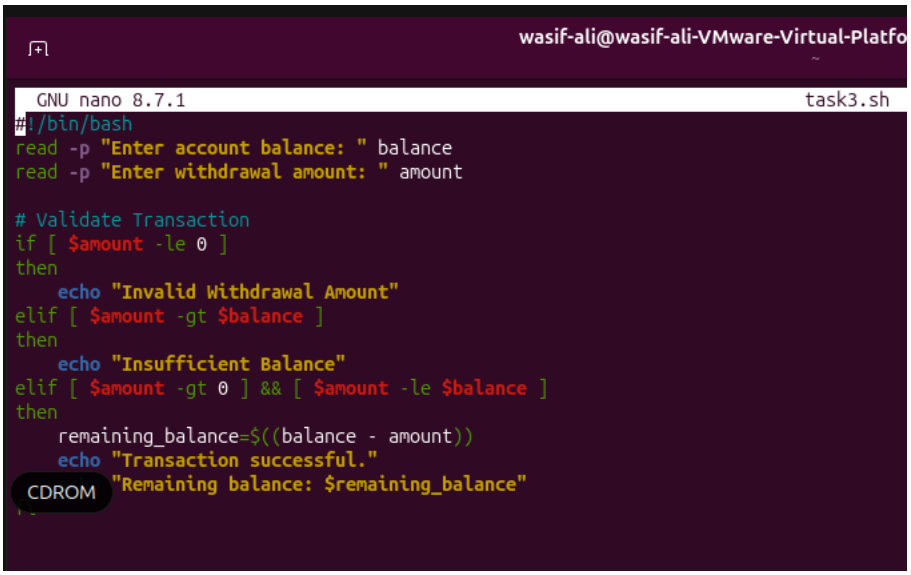
The script should:

- **Use** conditional statements to check whether the withdrawal amount is greater than zero.
- **Check** whether the withdrawal amount is within the available account balance.
- **Use** multiple conditions with && to validate a withdrawal.
- **Calculate** and **show** the remaining balance if the withdrawal is valid.
- **Report** Insufficient Balance if the withdrawal amount is greater than the available balance.
- **Report** Invalid Withdrawal Amount if the amount is zero or negative.

Lab Activity 3: ATM Transaction Validation

- The script takes the current account balance and a requested withdrawal amount as input.
- It validates that the withdrawal amount is greater than zero, rejecting the transaction as an "Invalid Withdrawal Amount" if it is zero or negative.
- It checks that the withdrawal does not exceed the available funds, reporting "Insufficient Balance" if it does.
- It calculates and displays the remaining account balance if the transaction is approved.

Script File (Shell Script Code)**:**



Commands:



Lab Activity 4 — Electricity Bill Calculator

Problem Statement

Produce a shell script that takes the number of electricity units consumed by a customer and **calculates** the total electricity bill.

The script should:

- **Use** conditional statements to identify the applicable electricity rate.
- **Calculate** the bill according to the following rates:

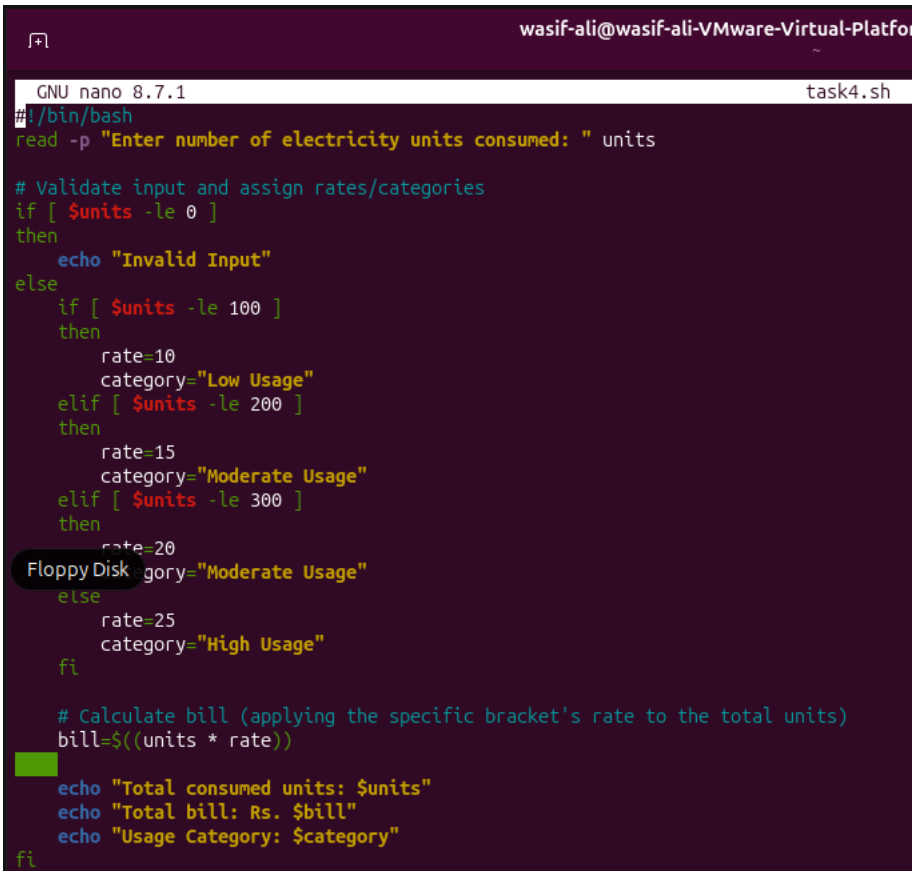
Units	Rate
1–100	Rs. 10/unit
101–200	Rs. 15/unit
201–300	Rs. 20/unit
Above 300	Rs. 25/unit

- **Show** the total number of consumed units.
- **Calculate** and **display** the total bill.
- **Relate** the number of consumed units with the appropriate usage category:
  - Up to 100 → Low Usage
  - 101–300 → Moderate Usage
  - Above 300 → High Usage
- **Report** Invalid Input if the number of units is zero or negative.
- **Demonstrate** the use of if-elif-else and arithmetic expressions in shell scripting.

Lab Activity 4: Electricity Bill Calculator

- The script takes the total number of consumed electricity units as input.
- It calculates the total bill based on progressive rates: Rs. 10/unit (1-100 units), Rs. 15/unit (101-200 units), Rs. 20/unit (201-300 units), and Rs. 25/unit (above 300 units).
- It assigns a usage category based on consumption: "Low Usage" (up to 100 units), "Moderate Usage" (101-300 units), or "High Usage" (above 300 units).
- It reports "Invalid Input" if the entered number of units is zero or negative

Script File (Shell Script Code)**:**



Commands:

